

FILIPP LOTSMANOV

Applied Data Science & AI Student
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EDUCATION

Breda University of Applied Sciences

BSc Applied Data Science & AI · 8.5/10 (≈3.8 GPA)

Sep 2024 – Present
Breda, Netherlands

PROFILE

Third-year Applied Data Science & AI student on the Data Scientist and Data Engineer tracks. I train vision, sequence and reinforcement learning models, measure them on held-out data, and deploy them for real-time and on-device inference.

TECHNICAL SKILLS

Languages: Python, Go, SQL
Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Stable Baselines3, Keras
Classical ML: scikit-learn, XGBoost, statsmodels
Computer vision: OpenCV, MediaPipe, torchvision, albumentations, U-Net
NLP & speech: SentencePiece/BPE, NLLB, faster-whisper, LangChain, FAISS
Optimisation & serving: TorchScript, quantisation, mixed-precision, FastAPI, Ollama, NVIDIA Jetson
MLOps & infra: MLflow, ClearML, Docker, Kubernetes, Azure, GitHub Actions, Airflow, Prometheus, Git, Portainer
Data & analysis: PostgreSQL, pandas, NumPy, GeoPandas, Shapely, Power BI

PROJECTS & TECHNICAL WORK

Autonomous Root Inoculation

Individual Project, BUas

Nov 2025 – Jan 2026

- CV and control system driving a lab robot to inoculate plant root tips: U-Net segmentation with geodesic root-length measurement cut error from 37.6% to 10.7% sMAPE (Kaggle private leaderboard).
- Benchmarked a PID controller (0.011–0.031 mm error) against a PPO agent (~99% at 1 mm); integrated pipeline hit 100% success across 50 targets at 0.646 mm mean accuracy.

Video Emotion Classification Pipeline

Group Project – Media Client

Sep – Nov 2025

- Nine-stage video-to-emotion-timeline system for an external client brief; fine-tuned DeBERTa-V2 on 428K samples to 0.8127 macro F1 held-out and 0.8865 on the external CARER benchmark.
- Built a Transformer encoder-decoder from scratch; owned the transcription and scene-detection stages.

NGT Sign Language Learning Platform

Group Project, BUas

Jan – Apr 2026

- Real-time browser app teaching the Dutch fingerspelling alphabet: ResidualMLP over 25 static classes at ~95% validation accuracy, BiLSTM for dynamic letters at ~92%.
- Served both models over a FastAPI WebSocket endpoint on MediaPipe hand landmarks, with per-letter model routing.

Concurrent ETL Pipeline in Go

Personal Project

May – Aug 2026

- Streaming CSV-into-PostgreSQL pipeline: 2.8–5.2 MB heap across a 2.3 GB, 27.6M-record file at ~45k rec/sec.
- 4.76× enrichment throughput via a goroutine worker pool; live-Postgres tests gate the Docker build and CI.

COMPETITIONS

BrabantHack 2026 – Winner, DEMCON Deep Tech Track

Team of 3

Apr 2026

- Predicted off-screen pedestrian positions from shadow imagery alone: PyTorch ResNet-50 backbone fused with 19 hand-crafted geometric features, reaching IoU 0.626 on the hidden test set.
- Iterated 0.52 → 0.626 IoU on ~11-min training runs, delivering a TorchScript + FastAPI demo in the hackathon window.

HERE Turn-Restriction Validation Challenge

Team of 3

May 2026

- Built an XGBoost classifier separating real from erroneous map turn restrictions from GPS probe traces, with GeoPandas/Shapely features in the Belgian national grid; 0.81 F1 under stratified 5-fold CV.
- Generated zero-match synthetic replicas of five proprietary HERE datasets for safe collaboration, and excluded synthetic training sets where labels leaked into route-derived violation features.

LANGUAGES

English (C2) · German (B1–B2) · Russian (B2) · Dutch (A2)